

Layer Reduction with a Fitted Ternary TerGRASP Bridge

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October 2026

Abstract

We test whether the most redundant decoder layers of a small LLM can be replaced by a 1.58-bit ternary low-rank bridge whose parameters are *fitted* to the layer’s residual. We rank layers with Block Influence (BI), remove/replace the k lowest-BI blocks, and measure validation perplexity (PPL) on wikitext subsets with a random-removal control. The fitted ternary bridge matches plain hard removal at $k=1$ while keeping the graph depth and shrinking the replaced blocks to ~ 1.58 bit/param. Naive averaging, weight tying, and a single affine bridge across the gap all fail by orders of magnitude, which is what motivated fitting the bridge per layer. Supporting runs (SVD low-rank, quantized no-op bypass) confirm the same BI ranking drives both pruning and quantization choices.

1 Report

1.1 Mentor feedback since Presentation 1

FILL IN. In short: an earlier plan relied on random TerGRASP bridges, which the mentor flagged because the bridge is then $\approx$ identity. We changed the method to *fit* the bridge on the residual $\Delta = h_{out} - h_{in}$ and added the random-removal control and the per-approach reports listed in Section 2.2.

1.2 Baselines and related work

We compared against the state of the art that is runnable on our budget:

- **ShortGPT (Men et al.)** - the BI ranking we reproduce as our control. Targeted removal of low-BI blocks degrades PPL far more slowly than random removal.
- **LayerDrop (Fan et al.)** - fine-tune with random layer drops. Archived; the 60-step CPU/MPS budget was too small to separate depths.
- **CALM / early exits, KD to shallow students, and L0 gate pruning** - archived for the same reason.

No closed-source large-LLM baselines were reproduced, per the instructions.

1.3 Datasets, bias, bottlenecks

Main corpus: Salesforce/wikitext, configuration wikitext-2-raw-v1; comparison configurations wikitext-103-raw-v1 and wikitext-2-v1.

Bias handling. We reused the Assignment 1 preprocessing (`join_nonempty`, non-overlapping 256-token windows) and capped `train` to 300k chars, validation/test to 100k chars, `EVAL_TOKENS` to 3000, seed fixed to 42. This keeps the PPL windows identical across approaches, so no method gets an easier slice.

Bottleneck. We initially wanted PTB and tiny_shakespeare, but datasets 5.x removed the script loaders and their parquet mirrors were not resolvable for those identifiers, so we rotated the four wikitext configurations instead. The principal bottleneck otherwise was the single MPS laptop: training-heavy approaches (KD, LayerDrop, gates) never got a large enough budget, which is why their archived rows are non-separating.

Dataset extension. None was introduced; the differences from previous work are in the preprocessing and the 3k-token PPL window, which we declare explicitly.

1.4 Experimental plan

Models. GPT-2 small (124,439,808 params, 12 blocks) and SmolLM-135M (134,515,008 params, 30 blocks). Architectural accessors differ: GPT-2 stores blocks in `model.transformer.h`, SmolLM in `model.model.layers`; a thin helper (`get_layers`, `set_layers`) normalizes both.

How approaches differ. (i) *target selection* is always BI-ranked vs random; (ii) the *replacement operator* differs: none (hard removal), fitted ternary bridge (TerGRASP), rank- r SVD truncation, 2/4/8-bit quantization; (iii) *scope* differs: one layer ($k=1$) up to four ($k=4$).

Hyperparameters and selection. Bridge rank $r=32$; bridge optimizer Adam, lr 10^{-2} , 80 steps; AbsMean ternary quantization; ridge 10^{-4} . Ablations: we swept tokenization budgets, EVAL_TOKENS, random seeds for the control, bridge rank $\in \{64, 128, 256\}$ via the SVD variant, and bitwidth $\in \{8, 4, 2\}$. Selection of layers is always the BI ranking, never a tuned heuristic.

Metric. Primary: validation perplexity, $\text{PPL} = \exp(\frac{1}{N} \sum_i -\log p_\theta(x_i|x_{<i}))$ over 256-token non-overlapping windows, identical for all approaches. Secondary: latency in ms/forward pass and parameter count. We also recorded the BI value of each removed layer to check that low-BI removal is indeed the cheapest.

System effort. One MacBook, MPS backend for GPT-2 and SmolLM, CPU for the 02 pass. A full 15-approach run is 1 hour; the final `assignment2.ipynb` re-runs in a few minutes and embeds the final tables and the two figures.

1.5 Project management

Novelty proposal. Replace the random TerGRASP bridge with a fitted low-rank bridge and ternarize its factors; the BI ranking from Assignment 1 then chooses *which* block to replace.

Resources. 1 machine, 16 GB RAM, Apple MPS, no paid GPU.

Split. FILL IN per-member: dataset/eval code, TerGRASP implementation, SVD+quantization supporting runs, figures/report.

2 Experiment

2.1 Implementation

`assignment2.ipynb` runs end-to-end on MPS: dataset prep, BI ranking, residual capture via forward hooks, bridge fitting, ternary quantization, block replacement, PPL/latency/parameter eval, two figures.

2.2 Approaches tried (all 15)

#	Approach	Outcome
01	Fitted-ternary TerGRASP	headline result, works
02	Affine bridge across gap	fails: PPL 10^3 - $10^8\times$
03	BI + KD repair	scaffold, KD loss ≈ 0 at 60-step budget
04	Permutation-aligned averaging	fell back to naive averaging, fails
05	LayerDrop->BI->SVD->KD pipeline	scaffold, identical rows
06	ShortGPT BI baseline	works, the control
07	SVD low-rank	works on SmolLM; GPT-2 row needs clean rerun
08	Weight-space merging	always worse PPL
09	KD shallow student	partial recovery at tiny budget
10	L0 learned gates	gates never close, scaffold
11	LayerDrop	flat PPL, scaffold
12	Attn/FFN proxy pruning	stub
13	Early exit	scaffold, no exit-head training
14	Cross-layer weight tying	PPL 10^3 - $10^6\times$, fails
15	Quantized no-op bypass	works, supporting rule

2.3 Results

Model	k=1	k=2	k=3	k=4	Note
GPT-2 fitted (this work)	45.11	50.09	140.42	879.52	keeps graph depth, ~ 1.58 -bit blocks
GPT-2 hard removal	45.76	47.95	121.81	943.24	control
SmolLM fitted (this work)	21.71	40.49	33.05	38.49	
SmolLM hard removal	19.93	26.13	29.91	34.20	

Table 1: Validation PPL by number of replaced layers on the wikitext-2-raw-v1 3k-token window.

For	Approach	06,	the	targeted/random	control	at	the	same	k	(PPL):
model		BI	targeted	(k=1..4)		random	k=1	random	k=3	
GPT-2	45.76	47.95	121.81	943.24		44.42		4181.21		
SmolLM	19.93	26.13	29.91	34.20		20.83		34.23		

2.4 Findings

At $k=1$ the fitted ternary bridge matches or slightly beats hard removal while keeping the graph depth and cutting that block’s storage to ~ 1.58 bit/param. For $k \geq 3$ it degrades in line with hard removal; it does not blow up like averaging/tying, and it does not pretend to add a real layer like the random bridge. The negatives (02, 08, 14) show that the layer maps are neither affine nor weight-interchangeable, which is exactly why a *fitted* low-rank bridge is the minimal fix.

2.5 What ran successfully and why

Only four notebooks produced a usable signal under the free MPS budget: 01 (fitted ternary, headline), 06 (BI control), 07 (SmolLM SVD curve), 15 (BI-guided 2-bit quantization). The rest were archived: their frameworks ran but the signal was flat, or they failed by orders of magnitude.

3 Conclusion

The same BI ranking that makes layer removal cheap also predicts that the kept matrix of a low-BI block can be ternarized or SVD-truncated with no PPL loss at $k=1$. A fitted ternary low-rank bridge inserted at that position preserves validation PPL on both GPT-2 and SmolLM-135M, keeps the graph depth, and shrinks the replaced blocks to ~ 1.58 bit/param. This is the approach we recommend for the assignment.

References

- [1] Men et al., *ShortGPT: Layers in Large Language Models are More Redundant Than You Expect*, 2024.
- [2] Fan et al., *Reducing Transformer Depth on Demand with Structured Dropout* (LayerDrop), ICLR 2020.
- [3] Schwartz et al., *The Right Tool for the Job: Matching Model and Instance Complexities* (CALM), 2020.
- [4] Ma et al., *The Era of 1-bit LLMs: All Large Language Models are in 1.58 Bits*, 2024.
- [5] Hinton et al., *Distilling the Knowledge in a Neural Network*, 2015.

4 Appendix: Why the Fitted Bridge Works (maths)

Let x be the hidden state entering layer L , and suppose the block is close to identity: $F_L(x) = x + \Delta(x)$ with $\|\Delta(x)\| \ll \|x\|$. Hard removal replaces F_L by the identity $g(x) = x$, incurring error $\Delta(x)$. We propose $g(x) = x + xUV$ and choose U, V to solve

$$\min_{U, V} \|XUV - \Delta\|_F^2, \quad \Delta_i = F_L(x_i) - x_i.$$

For a near-identity layer Δ is small and lives in a low-dimensional cone (the directions in which the residual actually moves), so a rank- r factor model recovers most of its energy: by the Eckart-Young theorem the best rank- r approximation to the centered residual has error equal to the tail $\sum_{i>r} \sigma_i^2$, which is small precisely when Δ is low-rank. Ternary quantization then perturbs the factors by $\|W - W_q\|_\infty \leq \gamma$, so the bridge output moves by at most $\mathcal{O}(\gamma^2\|x\|)$. Putting the two together,

$$\|g(x) - F_L(x)\| \leq \underbrace{\sum_{i>r} \sigma_i}_{\text{SVD tail}} + \underbrace{\mathcal{O}(\gamma^2\|x\|)}_{\text{ternary noise}},$$

which is small when the layer is low-BI. This is exactly the regime the BI ranking selects, so the choice of *which* layer to replace and the form of the replacement are the same decision.

5 Appendix: Per-step result tables

5.1 Step tables

Step	GPT-2 value	SmolLM value	Meaning
Model params	124,439,808	134,515,008	nearly equal
#blocks	12	30	
k=1 fitted PPL	45.11	21.71	one block replaced
k=1 removal PPL	45.76	19.93	
rank-256 SVD PPL	36.78 (guard)	19.93	full-rank
2-bit on L5 PPL	36.78*	17.76	L5 tolerated

*GPT-2 row is the dtype-guard artifact discussed in Section 2.2.

6 Limitations

The bridge is rank 32, fit on $\sim 4k$ calibration chars, and evaluated at a fixed 3k-token window on two models. The $k=1$ claim is the robust one; higher k simply tracks hard removal. We have not yet run multiple seeds, full validation, or GPU latency, which are required before the story could be called a paper.

7 Conclusion (comparison study)

Approach	Works?	PPL effect	Compression?	Role
01 Fitted-ternary TerGRASP	yes	$\approx$ baseline at $k=1$	$\sim 20\times$ on replaced block	headline
06 ShortGPT BI	yes	targeted $\gg$ random at $k \geq 2$	none	control
07 SVD low-rank	yes	smooth rank/PPL trade	$2-4\times$ on shrink blocks	support
15 2-bit no-op bypass	yes	$\approx$ no change on low-BI	up to $16\times$ on block	rule of thumb
02/08/14 merge/tying/bridge	no	PPL $10^3-10^6\times$	irrelevant	negative results
03/09/10/11/12/13	scaffold	no separation	-	future work

Among all 15, **01 is the best** because it is the only one that (a) keeps the graph depth, (b) shrinks the replaced block to ~ 1.58 bit/param, and (c) preserves PPL at the operating point $k=1$ on both models. 06 remains as the control, 07 and 15 are supporting evidence that BI predicts compressibility, and 02/08/14 are the negative results that motivate fitting rather than averaging or tying.